

Permuted Binary Strings

Time limit: 1.00 s

Memory limit: 512 MB

There is a hidden permutation $a_1, a_2, \dots, a_n$ of integers $1, 2, \dots, n$. Your task is to find this permutation. To do this, you can ask questions: you can choose a binary string $b_1 b_2 \dots b_n$ and you will receive the binary string $b_{a_1} b_{a_2} \dots b_{a_n}$.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n : the length of the permutation.

On your turn, you can print one of the following:

- " $? b_1 b_2 \dots b_n$ ", where $b_i \in \{0, 1\}$: The grader will return the binary string $b_{a_1} b_{a_2} \dots b_{a_n}$.
- " $! a_1 a_2 \dots a_n$ ": report that the hidden permutation is $a_1, a_2, \dots, a_n$. Your program must terminate after this.

Each line should be followed by a line break. You must make sure the output gets flushed after printing each line.

Constraints

- $1 \leq n \leq 1000$
- you can ask at most 10 questions of type $?$

Example

```
3
? 100
100
? 010
001
? 001
010
! 1 3 2
```

Explanation: The hidden permutation is $[1, 3, 2]$. In the first question $b_1 b_2 b_3 = 100$ and the grader returns $b_{a_1} b_{a_2} b_{a_3} = b_1 b_3 b_2 = 100$. In the second question $b_1 b_2 b_3 = 010$ and the grader returns $b_1 b_3 b_2 = 001$.